

Mathematical Analysis Zorich Comprehensions

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21st August, 2025

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Figure 1: One week, one Zorich

1 Chapter2. The Real Numbers

1.1 Comprehensions

The main content covers the basic properties of real numbers, the definitions and properties of rational and irrational numbers, natural numbers and mathematical induction, and, most importantly, several equivalence principles describing the completeness of real numbers (more generally, the density of real numbers). (Zorich calls these fundamental lemmas, which serve as the foundation for constructing real number theory, but calling them principles is fine.) Finally, some information on Cantor's set theory is added.

The method of describing the completeness of real numbers using Dedekind partitioning, which we learned in Fehkingoltz's "Calculus Tutorial (I)" can also be incorporated into this.

Basic Lemmas of the Real Number Completeness Theorem

1. (The least upper (lower) bound principle) Every non-empty set of real numbers that is bounded from above (below) has a unique upper (lower) bound.
2. (Cauchy-Cantor Principle) (The nested interval lemma) The set of closed intervals on the real number axis whose length $\rightarrow 0$ determines a unique point (a common point that belongs to all these closed intervals).
3. (Borel-Lebesgue Principle) (The finite covering lemma) Every system of open intervals covering a closed interval contains a finite subcovering.
4. (The Bolzano-Weierstrass Principle) (The limit point lemma) Every bounded infinite set of real numbers has at least one limit point.

Notes. 1. If the length of the closed intervals is not approaching 0, there are still existing common points, but they are not unique.
2. If infinite is limit point, then the Bolzano-Weierstrass Principle is suit for unbounded infinite set of real numbers.

2 important properties of natural numbers $\mathbb{Q}$.

2 Important Properties of Natural Numbers

Thm1. (The Fundamental Theorem of Arithmetic) $\forall n \in \mathbb{N}$, it could be represented uniquely as: $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, where p_j are different prime numbers, α_j are positive integers.

Thm2. (The Principle of Archimedes) Given $h > 0$, then $\forall x \in \mathbb{R}$, $\exists$ unique $k \in \mathbb{Z}$, s.t. $(k-1)h \leq x \leq kh$.

Cantor Set Theory.

Countable and Uncountable Sets

Proposition 1. $\text{card}(\text{Countable Set}) = \text{card}(\mathbb{N}) = \aleph_0$, $\text{card}(\text{Uncountable Set}) = \text{card}(\mathbb{R}) = \aleph_1$ (cardinality of the continuum).

Proposition 2. Any subset of a at most countable set ($\text{card}(X) \leq \text{card}(\mathbb{N})$) is at most countable; the at most countable union of at most countable sets is at most countable.

Proposition 3. (Corollary) The direct sum (or direct product) of countable sets is countable. (Corollary) The set of natural numbers or algebraic numbers is countable.

Thm 1. (Cantor's Theorem) $\text{card}(\mathbb{N}) < \text{card}(\mathbb{R})$.

Proposition 4. (Corollary) the irrational numbers or transcendental numbers exist, and the set of all of them is uncountable.

1.2 Problems Sets

An interesting example-the Cantor set

Example. Let $x \in [0, 1] \subset \mathbb{R}$, its ternary expansion is $x = 0.a_1a_2a_3 \cdots$ ($a_i \in \{0, 1, 2\}$) satisfying: $\forall i \in \mathbb{N}, a_i \neq 1$, then the set of all such elements is called the Cantor set.

Notes. The Cantor set is uncountable (It's easy: the elements of it can be onto mapped to all binary decimals on $[0, 1]$), and we'll then learn: Cantor set is measure zero in sense of Lebesgue.

2 2 Classical Limits

2.1 Original

2.1.1 $\lim_{n \rightarrow \infty} \frac{a^n}{n^k}$

The problem is:

$$\lim_{n \rightarrow \infty} \frac{a^n}{n^k} \quad \text{where } a > 1, k > 0$$

Solution (Solution1). Based on **Binomial Theorem**
 $a^n = [a \triangleq 1 + \lambda] = 1 + n\lambda + \frac{n(n-1)}{2}\lambda^2 + \dots > \frac{n(n-1)}{2}\lambda^2$
 also $n-1 > \frac{n}{2}$, as $n > 2$, then $a^n > \frac{n^2}{4}(a-1)^2$.
 As $k = 1$, $\lim_{n \rightarrow \infty} \frac{a^n}{n} = \infty$, and as $k > 1$, $\lim_{n \rightarrow \infty} \frac{a^n}{n^k} = \lim_{n \rightarrow \infty} \left(\frac{(a^{\frac{1}{k}})^n}{n}\right)^k = \infty$.

Solution (Solution2). Based on **L'Hospital's Rule**
 $\lim_{n \rightarrow \infty} \frac{a^n}{n^k} = \lim_{n \rightarrow \infty} \frac{\ln a \cdot a^n}{k \cdot n^{k-1}} = \lim_{n \rightarrow \infty} \frac{(\ln a)^2 \cdot a^n}{k(k-1) \cdot n^{k-2}} = \dots =$
 $\lim_{n \rightarrow \infty} \frac{(\ln a)^k \cdot a^n}{k!} = \infty$.

Solution (Solution3). (Haven't found a totally different solution yet.)

2.1.2 $\lim_{n \rightarrow \infty} n^{\frac{1}{n}}$

The problem is:

$$\lim_{n \rightarrow \infty} n^{\frac{1}{n}}$$

Solution (Solution1). Due to 1. $a^n > \frac{n^2}{4}(a-1)^2$, let $a = n^{\frac{1}{n}}$, then $0 < n^{\frac{1}{n}} < \frac{2}{\sqrt{n}}$. Thus $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$.

Solution (Solution2). Based on Mean Value Inequalities Chain(GM-AM): $n^{\frac{1}{n}} = (\sqrt{n} \cdot \sqrt{n} \cdot 1 \dots 1)^{\frac{1}{n}} \leq \frac{2\sqrt{n} + n - 2}{n}$, then $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$.

2.2 Some repurposed variants

2.2.1 $\lim_{n \rightarrow +\infty} \sqrt[n]{n!}$

The problem is:

$$\lim_{n \rightarrow +\infty} \sqrt[n]{n!}$$

Solution (Solution1). Based on **Stirling's Approximation**: $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$, then $\sqrt[n]{n!} \sim \sqrt[n]{\sqrt{2\pi n} \left(\frac{n}{e}\right)^n} = \frac{n}{e} \cdot (2\pi n)^{\frac{1}{2n}}$. Thus $\lim_{n \rightarrow +\infty} \sqrt[n]{n!} = \infty$.

Solution (Solution2). $n! \geq \left(\frac{n}{2}\right)^{\frac{n}{2}}$ because half of the factors are at least $\frac{n}{2}$

Solution (Solution3). Consider Taylor series, $e^x \geq \frac{x^n}{n!}$, $\forall x \geq 0, n \in \mathbb{N}$, then $n! \geq \frac{x^n}{e^x}$, take $x = n$ problem solved.

Solution (Solution4). Based on **Stolze Cesaro Theorem**:
 $\lim_{n \rightarrow +\infty} \sqrt[n]{n!} = \lim_{n \rightarrow +\infty} \frac{n!^{\frac{1}{n}}}{1} = \lim_{n \rightarrow +\infty} \frac{\ln(n!)}{n} = \lim_{n \rightarrow +\infty} \frac{\ln(n) + \ln(n-1) + \dots + \ln(1)}{n}$,
then apply Stolze Cessaro Theorem, $\lim_{n \rightarrow +\infty} \frac{\ln(n) + \ln(n-1) + \dots + \ln(1)}{n} =$
 $\lim_{n \rightarrow +\infty} \frac{\ln(n+1)}{1} =$, then $\lim_{n \rightarrow +\infty} \sqrt[n]{n!} = \infty$.

Solution (Solution5). Using a multiplicative variant of Gauss' trick we get: $(n!)^2 = (1 \cdot n)(2 \cdot (n-1)) \dots ((n-1) \cdot 2)(n \cdot 1) \geq n^n$.

3 Multiple Integrals Short Sheet (Calculus)

3.1 Development of 1-dim Reimann Integral-Line Integral

3.1.1 Intrduction (ds)

Introduction

$$I = \int_C f(x, y) ds, \quad C : r = r(\theta), \quad \theta \in [\theta_1, \theta_2]$$

$$C : \begin{cases} x = r(\theta) \cos \theta \\ y = r(\theta) \sin \theta \end{cases}, \quad f(x, y) = f(r \cos \theta, r \sin \theta)$$

$$ds = \sqrt{x'(\theta)^2 + y'(\theta)^2} d\theta = \sqrt{r'^2 + r^2} d\theta$$

$$I = \int_{\theta_1}^{\theta_2} f(r \cos \theta, r \sin \theta) \sqrt{r'^2 + r^2} d\theta$$

3.1.2 I-Line Integral

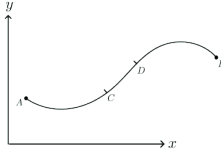
Definition

$$I = \int_C f(x, y) ds = \int_{D_t} f(\varphi(t), \psi(t)) \sqrt{\varphi'(t)^2 + \psi'(t)^2} dt$$

A mean value property

$$\rho(x, y) = f(x, y), \quad \int_{C(A, B)} f(x, y) ds = M, \quad \int_{C(C, D)} f(x, y) ds = m$$

$$\frac{\int_{C(C, D)} f(x, y) ds}{\int_{C(A, B)} f(x, y) ds} = \frac{\int_{C(C, D)} C ds}{\int_{C(A, B)} C ds} = \frac{S_{CD}}{S_{AB}}$$



3.1.3 II-Line Integral

Definition

$$I = \int_C f(x, y) dx = [x = \varphi(t), y = \psi(t)] = \int_{D_t} f(\varphi(t), \psi(t)) \varphi'(t) dt$$

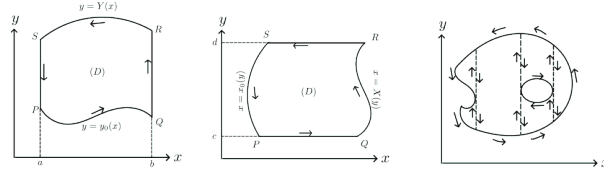
$$I = \int_C f(x, y) dy = [x = \varphi(t), y = \psi(t)] = \int_{D_t} f(\varphi(t), \psi(t)) \psi'(t) dt$$

$$I = \int_C P dx + Q dy = [x = \varphi(t), y = \psi(t)] = \int_{D_t} [P\varphi' + Q\psi'] dt$$

3.1.4 Calculate area using II-Line Integral

Calculate

$D = - \int_C y dx = \int_C x dy = \frac{1}{2} \int_C x dy - y dx$, where C is a simple closed curve with positive direction.



3.1.5 Conditions for curve integrals to be independent of the path

Theorem

In simply connected space D and given a path $C : A \rightarrow B$, II-line integral $\int_C P dx + Q dy$ is dependent of the integration path $C : A \rightarrow B$ i.f.f. $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

Proof idea

Necessity " $\Rightarrow$ ":

By studying the differentiation method (partial differentiation with respect to x and y) of $F(x, y) = \int_{(x_0, y_0)}^{(x, y)} P dx + Q dy$ and the primitive function of $P dx + Q dy$ (regarding it as the total differential of some function) (i.e., $\exists \Phi$ such that $P = \frac{\partial \Phi}{\partial x}$, $Q = \frac{\partial \Phi}{\partial y}$ and $\int_C P dx + Q dy = \Phi(x, y)|_{(x_A, y_A)}^{(x_B, y_B)}$), we can conclude that the necessary and sufficient condition for $\int_C P dx + Q dy$ to be path-independent is that $P dx + Q dy (= d\Phi(x))$ is an exact differential. The latter implies $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ (necessity).

Sufficiency " $\Leftarrow$ ":

Fix $y \in [c, d]$, integrate $P = \frac{\partial \Phi}{\partial x}$ on $[x_0, x] \subset D$, then $\Phi(x, y) = \int_{x_0}^x P dx + \Phi(x_0, y)$. Also, fix $x = x_0$, integrate $Q = \frac{\partial \Phi}{\partial y}$ on $[y_0, y] \subset D$, then $\Phi(x_0, y) = \int_{y_0}^y Q(x_0, y) dy + \Phi(x_0, y_0)$.

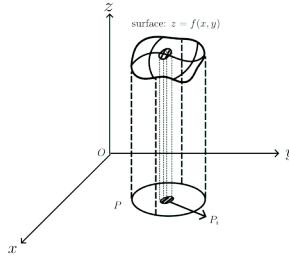
Then $\Phi(x, y) = \int_{x_0}^x P(x, y) dx + \int_{y_0}^y Q(x_0, y) dy + \Phi(x_0, y_0)$ or $\Phi(x, y) = \int_{y_0}^y Q(x, y) dy + \int_{x_0}^x P(x, y_0) dx + \Phi(x_0, y_0)$, which satisfies $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

3.2 2-dim Multiple Integral

3.2.1 Definition

Definition

$$\iint_P f(x, y) dP = \lim_{\lambda = \max_i \{P_i\} \rightarrow 0} \sum_i f(\xi_i, \eta_i) \Delta P_i = V$$



3.2.2 Calculate: 2-dim Multiple Integral $\rightarrow$ Iterated Integral

2-dim Multiple Integral $\rightarrow$ Iterated Integral

Rectangle $P = [a, b] \times [c, d]$:

If $\iint_P f(x, y) dP$ exists and $I_1 = \int_a^b f(x, y) dx$ exists $\forall y \in [c, d]$, then $\iint_P f(x, y) dP = \int_c^d dy \int_a^b f(x, y) dx$.

If $\iint_P f(x, y) dP$ exists and $I_2 = \int_c^d f(x, y) dy$ exists $\forall x \in [a, b]$, then $\iint_P f(x, y) dP = \int_a^b dx \int_c^d f(x, y) dy$.

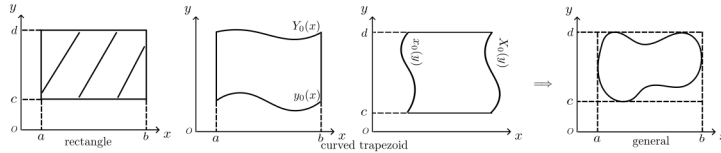
$\Rightarrow$ If I_1 and I_2 exist, then $\iint_P f(x, y) dP = \int_a^b dx \int_c^d f(x, y) dy = \int_c^d dy \int_a^b f(x, y) dx$.

Curved trapezoid $P = [a, b] \times [y_0(x), Y(x)]$ or $P = [x_0(y), X(y)] \times [c, d]$:

If $\iint_P f(x, y) dP$ exists and $I_2 = \int_{y_0(x)}^{Y(x)} f(x, y) dy$ exists $\forall x \in [a, b]$, then $\iint_P f(x, y) dP = \int_a^b dx \int_{y_0(x)}^{Y(x)} f(x, y) dy$.

If $\iint_P f(x, y) dP$ exists and $I_1 = \int_{x_0(y)}^{X(y)} f(x, y) dx$ exists $\forall y \in [c, d]$, then $\iint_P f(x, y) dP = \int_c^d dy \int_{x_0(y)}^{X(y)} f(x, y) dx$.

$\Rightarrow$ If I_1 and I_2 exist, then $\iint_P f(x, y) dP = \int_a^b dx \int_{y_0(x)}^{Y(x)} f(x, y) dy = \int_{y_0(x)}^{Y(x)} dy \int_a^b f(x, y) dx$.



3.2.3 Green's Formula: 2-dim Multiple Integral $\longleftrightarrow$ II-Line Integral

Green's Formula

$$\iint_D \left(\frac{\partial P}{\partial x} - \frac{\partial Q}{\partial y} \right) dx dy = \oint_{C=\partial D} P dy + Q dx$$

or

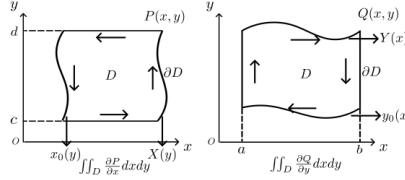
$$\iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx dy = \oint_{C=\partial D} P dy - Q dx$$

where D is a simply connected domain in $\mathbb{R}^2$ with boundary $C = \partial D$ and P, Q are continuously differentiable functions on D .

Proof idea

$P, Q, \frac{\partial P}{\partial x}, \frac{\partial Q}{\partial y}$ continuously on D , $\partial D \triangleq C$.

$$\begin{aligned} \iint_D \frac{\partial P}{\partial x} dx dy &= \int_c^d dy \int_{x_0(y)}^{X(y)} \frac{\partial P}{\partial x} dx = \int_c^d [P(X(y), y) - P(x_0(y), y)] dy = \\ &= \oint_{\partial D} P(x, y) dy \\ \iint_D \frac{\partial Q}{\partial y} dx dy &= \int_a^b dx \int_{y_0(x)}^{Y(x)} \frac{\partial Q}{\partial y} dy = \int_a^b [Q(x, Y(x)) - Q(x, y_0(x))] dx = \\ &= - \oint_{\partial D} Q(x, y) dx \end{aligned}$$



3.2.4 Variable Substitution in Double Integrals

Intro

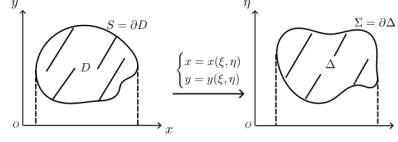
$$D = \oint_{S=\partial D} x dy = \left[\begin{matrix} x = x(\xi, \eta) \\ y = y(\xi, \eta) \end{matrix} \right] = \oint_{\Sigma=\partial \Delta} x(\xi, \eta) \frac{\partial y}{\partial \xi} d\xi + x(\xi, \eta) \frac{\partial y}{\partial \eta} d\eta = \iint_{\Delta} \frac{D(x, y)}{D(\xi, \eta)} d\xi d\eta$$

Analysis

As we could see, the essential of $\begin{cases} x = x(\xi, \eta) \\ y = y(\xi, \eta) \end{cases}$ actually is coordinate transform between the x, y -plane and the ξ, η -plane.

$$\begin{aligned} D = \oint_{\Sigma} x dy &= \left[\begin{matrix} x = x(\xi, \eta) \\ y = y(\xi, \eta) \end{matrix}, \begin{matrix} \xi = \xi(t) \\ \eta = \eta(t) \end{matrix} \right] = \int_{\alpha}^{\beta} x(\xi(t), \eta(t)) \left(\frac{\partial y}{\partial \xi} \xi'(t) + \frac{\partial y}{\partial \eta} \eta'(t) \right) dt \\ &= \oint_{\Sigma} x(\xi, \eta) \frac{\partial y}{\partial \xi} d\xi + x(\xi, \eta) \frac{\partial y}{\partial \eta} d\eta, \quad \text{let } x(\xi, \eta) \frac{\partial y}{\partial \xi} \triangleq P(x, y), \quad x(\xi, \eta) \frac{\partial y}{\partial \eta} \triangleq Q(x, y), \quad \text{then } D = \oint_{\Sigma} P d\xi + Q d\eta = \left(\frac{\partial P}{\partial \eta} - \frac{\partial Q}{\partial \xi} \right) \iint_{\Delta} d\xi d\eta = \iint_{\Delta} \frac{D(x, y)}{D(\xi, \eta)} d\xi d\eta. \end{aligned}$$

Figures for aforementioned analysis



Idea of Analysis:

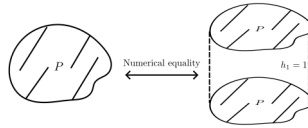
Using $\begin{cases} x = x(\xi, \eta) = x(\xi(t), \eta(t)) \\ y = y(\xi, \eta) = y(\xi(t), \eta(t)) \end{cases}$ (coordinate transform with parameterization), transform the II-line integration $\oint_S xdy$ (or $-\oint_S ydx$, or $\frac{1}{2} \oint_S xdy - ydx$) on x, y -plane to the II-line integral $\oint_{\Sigma} P(\xi, \eta)d\xi + Q(\xi, \eta)d\eta$, then using Green's formula to transform it to double integral $\iint_{\Delta} (\frac{\partial Q}{\partial \xi} - \frac{\partial P}{\partial \eta})d\xi d\eta$ on ξ, η -plane, following is how we calculate the double integral (often transform it to iterated integral $\int_{D_{\xi}} d\xi \int_{\eta_0(\xi)}^{\eta_1(\xi)} (\frac{\partial Q}{\partial \xi} - \frac{\partial P}{\partial \eta})d\xi$, or others).

/Shortly, calculate the area of D on x, y plane: coordinate transform with parameterization $\rightarrow$ Green's formula $\rightarrow$ Double integration calculate/

Comparison:

Recaping the previous $D = \frac{1}{2} \oint_S xdy - ydx$, the original calculation method for this II-line integral is that find thier parametric equation $\begin{cases} x = x(t) \\ y = y(t) \end{cases}, t \in [a, b]$ for different (closed) curve S , then substitute into the calculation. From here, we could see that the essence of II-line integral is definite integral and has chirality.

However, "coordinate transform + Green's formula" method: $D = \iint_{\Delta} \frac{D(x, y)}{D(\xi, \eta)} d\xi d\eta$ as long as we master the coordinate transformation of a type of shape equation, we can easily calculate the results. Although there is a parameterization in the derivation, it is not reflected in the result (it can be seen from the process that if the parameterization is continued, it will be the result of the II-line integral in ξ, η -plane). In essence, the area is calculated as the volume with the same value, similiar with $\frac{1}{2} \oint_S xdy - ydx = \iint_D 1dxdy$ or $\oint_S xdy = \iint_D 1dxdy$ (If you unfold the left and right sides under the curved trapezoid, you can find the consistency.), it's just that the physical meanings of the two equations are different.



/Note: The choice of which method should be based on which is more convenient, parametric equations or coordinate transformations./

Variable Substitution in Double Integral

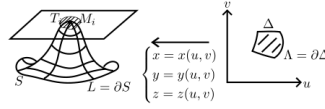
$$\iint_D f(x, y) dx dy = \begin{cases} x = x(\xi, \eta) \\ y = y(\xi, \eta) \end{cases} = \iint_{\Delta} f(x(\xi, \eta), y(\xi, \eta)) \left| \frac{D(x, y)}{D(\xi, \eta)} \right| d\xi d\eta$$

3.3 Surface Integral

3.3.1 Definition and basics

Definition

$S = \lim_{\lambda \rightarrow 0} \sum_i T_i$, where T_i is the projection area of S_i on the tangent plane at point $M_i \in S_i$.



Calculate Surface Integral (Omit matrix-vector operations)

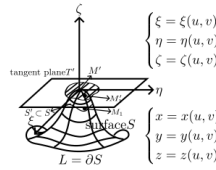
$$S = \lim_{\lambda \rightarrow 0} \sum_i T_i = \iint_{\Delta} \sqrt{A^2 + B^2 + C^2} du dv = \iint_{\Delta} \sqrt{EG - F^2} du dv$$

where $A = \begin{vmatrix} y'_u & z'_u \\ y'_v & z'_v \end{vmatrix}$, $B = \begin{vmatrix} z'_u & x'_u \\ z'_v & x'_v \end{vmatrix}$, $C = \begin{vmatrix} x'_u & y'_u \\ x'_v & y'_v \end{vmatrix}$, $E = \sum_{x,y,z} x'^2_u$, $F = \sum_{x,y,z} x'_u x'_v$, $G = \sum_{x,y,z} x'^2_v$.

Procedure:

$\forall M_1 \in S'$, the coordinates of its projection M'_1 on T' are $\begin{cases} \xi = \xi(u, v) \\ \eta = \eta(u, v) \\ \zeta = \zeta(u, v) \end{cases}$,

then $S'_T = \iint_{\Delta} \left| \frac{D(\xi, \eta)}{D(u, v)} \right| du dv$, then the results could appear after a series of matrix-vector operations.



3.3.2 Surface Integral

I-Surface Integral

$$\begin{aligned}
 I &= \iint_S f(x, y, z) \, ds \\
 &= [x = x(u, v), y = y(u, v), z = z(u, v), /parameterization/ \\
 &\quad dS = \sqrt{A^2 + B^2 + C^2} \, du \, dv = \sqrt{EG - F^2} \, du \, dv] \\
 &= \iint_{\Delta} f(x(u, v), y(u, v), z(u, v)) \sqrt{A^2 + B^2 + C^2} \, du \, dv \\
 &= \iint_{\Delta} f(x(u, v), y(u, v), z(u, v)) \sqrt{EG - F^2} \, du \, dv
 \end{aligned}$$

Special case:

If $u = x, v = y, z = z(x, y)$,

$$\begin{aligned}
 I &= \iint_{\Delta} f(x, y, z(x, y)) \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \, dx \, dy \\
 &= \iint_{\Delta} f(x, y, z(x, y)) \frac{dx \, dy}{|\cos(v)|}
 \end{aligned}$$

II-Surface Integral

$$\begin{aligned}
 I &= \iint_S f(x, y, z) \, dx \, dy = \text{or} \iint_S f(x, y, z) \, dy \, dz = \text{or} \iint_S f(x, y, z) \, dz \, dx \\
 &\text{or} \iint_S P \, dy \, dz + Q \, dz \, dx + R \, dx \, dy
 \end{aligned}$$

Connection between I and II-Surface Integrals

$$\begin{aligned}
 \iint_S f(x, y, z) \, dx \, dy &= \iint_S f(x, y, z) \cos(v) \, ds \\
 \text{(II)} &\qquad \qquad \text{(I)} \\
 &\text{(Calculate it: parameterize to get double integral)} \\
 &= [x = x(u, v), y = y(u, v), z = z(u, v), /parameterize/ \\
 &\quad \cos(v) = \pm \frac{C}{\sqrt{A^2 + B^2 + C^2}}, \, dS = \sqrt{A^2 + B^2 + C^2} \, du \, dv] \\
 &= \pm \iint_{\Delta} f(x(u, v), y(u, v), z(u, v)) \cdot C \, du \, dv
 \end{aligned}$$

or

$$\underbrace{\iint_S P dy dz + Q dz dx + R dx dy}_{(II)} = \underbrace{\iint_S P \cos(\lambda) + Q \cos(\mu) + R \cos(\nu) dS}_{(I)}$$

(Calculate it: parameterize to get double integral)

$$= \left[\begin{cases} x = x(u, v) \\ y = y(u, v) \\ z = z(u, v) \end{cases} \quad / \text{parameterize}/ \right.$$

$$\cos(\lambda) = \pm \frac{A}{\sqrt{A^2 + B^2 + C^2}}, \dots,$$

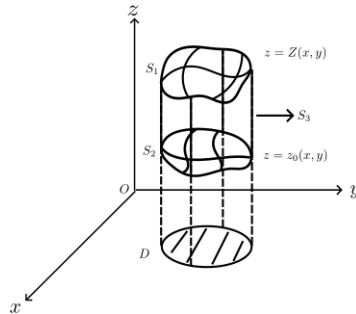
$$dS = \sqrt{A^2 + B^2 + C^2} du dv]$$

$$= \pm \iint_{\Delta} [P \cdot A + Q \cdot B + R \cdot C] du dv$$

3.3.3 Calculate Volume using Surface Integral

Calculate Volume using Surface Integral

$$\begin{aligned} V &= \iint_D Z(x, y) dx dy - \iint_D z_0(x, y) dx dy \\ &= \iint_{S_2} z dx dy + \iint_{S_1} z dx dy \\ &= \iint_S z dx dy \quad (S = S_1 + S_2 + S_3, \text{ outside}) \\ \text{or} &= \iint_S x dy dz \\ \text{or} &= \iint_S y dz dx \\ \text{or} &= \frac{1}{3} \iint_S (x dy dz + y dz dx + z dx dy) \end{aligned}$$



3.3.4 Stoke's Formula: II-Surface Integral $\longleftrightarrow$ II-Line Integral

Stoke's Formula $\Leftarrow$ Development of Green's Formula

$$\int_{L=\partial S} P dx + Q dy + R dz = \iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy + \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dy dz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dz dx$$

Proof idea

Using Green's formula on LHS as parameterizing $[x = x(u, v), y = y(u, v), z = z(u, v)]$, it's easy to prove.

3.4 3-dim Multiple Integral

3.4.1 Definition and basics

Definition

$$\iiint_V f(x, y, z) dV = \lim_{\lambda \rightarrow 0} \sum_i f(x_i, y_i, z_i) \Delta V_i.$$

Calculate

$$\begin{aligned} \iiint_V f(x, y, z) dV &= \int_a^b dx \iint_{P_y z} f(x, y, z) dy dz \\ &\quad \left(\iint_{P_y z} f(x, y, z) dy dz \exists \right) \\ &= \int_a^b dx \int_c^d dy \int_e^f f(x, y, z) dz \\ &\quad \left(\int_e^f f(x, y, z) dz \exists \right) \\ &\text{also} = \iint_{P_x y} dx dy \int_e^f f(x, y, z) dz \\ \iiint_V f(x, y, z) dV &= \int_{x_0}^X dx \iint_{P_y z} f(x, y, z) dy dz \\ &\quad \left(\iint_{P_y z} f(x, y, z) dy dz \exists \right) \\ &= \int_{x_0}^X dx \int_{y_0}^{Y(x)} dy \int_{z_0(x, y)}^{Z(x, y)} f(x, y, z) dz \\ &\quad \left(\int_{z_0(x, y)}^{Z(x, y)} f(x, y, z) dz \exists \right) \\ &\text{also} = \iint_{P_x y} dx dy \int_{z_0(x, y)}^{Z(x, y)} f(x, y, z) dz \end{aligned}$$

3.4.2 Gauss's Formual 3-dim Multiple Integral $\longleftrightarrow$ II-Surface Integral

Gauss's Formula: $\Leftarrow$ Dimension-increasing version of Stokes' formula

$$\begin{cases} \iiint_V \frac{\partial R}{\partial z} dx dy dz = \iint_{S=\partial V} R dx dy \\ \iiint_V \frac{\partial P}{\partial x} dy dz dx = \iint_{S=\partial V} P dy dz \\ \iiint_V \frac{\partial Q}{\partial y} dz dx dy = \iint_{S=\partial V} Q dz dx \end{cases}$$

$$\Sigma : \implies \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx dy dz = \iint_{S=\partial V} P dy dz + Q dz dx + R dx dy \quad (\text{S is along the outside})$$

3.4.3 Variable Substitution in 3-dim Multiple Integral

Intro

$$\begin{aligned} V &= \iint_S z dx dy \\ &(\text{coordinate transform}) \\ &= \iint_E z \cdot \frac{D(x, y)}{D(u, v)} du dv \\ &= \iint_E z \cdot \left[\frac{D(x, y)}{D(\xi, \eta)} \frac{D(\xi, \eta)}{D(u, v)} + \frac{D(x, y)}{D(\eta, \zeta)} \frac{D(\eta, \zeta)}{D(u, v)} + \frac{D(x, y)}{D(\zeta, \xi)} \frac{D(\zeta, \xi)}{D(u, v)} \right] du dv \\ &(\text{II-Surface Integral}) \quad (\text{using Green's formula}) \\ &= \iint_{\Sigma} z \frac{D(x, y)}{D(\xi, \eta)} d\xi d\eta + z \frac{D(x, y)}{D(\eta, \zeta)} d\eta d\zeta + z \frac{D(x, y)}{D(\zeta, \xi)} d\zeta d\xi \\ &= \iiint_{\Delta} \left(\frac{\partial}{\partial \zeta} \left[z \frac{D(x, y)}{D(\xi, \eta)} \right] + \frac{\partial}{\partial \xi} \left[z \frac{D(x, y)}{D(\eta, \zeta)} \right] + \frac{\partial}{\partial \eta} \left[z \frac{D(x, y)}{D(\zeta, \xi)} \right] \right) d\xi d\eta d\zeta \\ &(\text{rearrange}) \\ &= \iint_{\Delta} \left| \frac{D(x, y, z)}{D(\xi, \eta, \zeta)} \right| d\xi d\eta d\zeta \end{aligned}$$

Variable Substitution in 3-dim Multiple Integral

$$\begin{aligned} &\iiint_D f(x, y, z) dx dy dz \\ &= \left[\begin{cases} x = x(\xi, \eta, \zeta) \\ y = y(\xi, \eta, \zeta) \\ z = z(\xi, \eta, \zeta) \end{cases} \quad /parameterize/ \right] \\ &= \iiint_{\Delta} f(x(\xi, \eta, \zeta), y(\xi, \eta, \zeta), z(\xi, \eta, \zeta)) \cdot \left| \frac{D(x, y, z)}{D(\xi, \eta, \zeta)} \right| d\xi d\eta d\zeta \end{aligned}$$

3.5 n -dim Multiple Integral

3.5.1 Definition and basics

Definition

$$\int_I f(x^1, \dots, x^n) dx^1 \dots dx^n = \lim_{\lambda(P) \rightarrow 0} \sigma(f, P, \xi)$$

where I is an n -dimensional interval, P is a partition of I , $\lambda(P)$ is the measure of P , and $\sigma(f, P, \xi)$ is the Riemann sum.

Calculate

$$\int_I f(x^1, \dots, x^n) dx^1 \dots dx^n$$

(Special)

$$= \int_{a_1}^{b_1} dx^1 \int_{a_2}^{b_2} dx^2 \dots \int_{a_n}^{b_n} f(x^1, \dots, x^n) dx^n$$

(General)

$$= \int_{x_1}^{X_1} dx^1 \int_{x_2(x_1)}^{X_2(x_1)} dx^2 \dots \int_{x_n(x_1, \dots, x_{n-1})}^{X_n(x_1, \dots, x_{n-1})} f(x^1, \dots, x^n) dx^n$$

3.5.2 Variable Substitution in n -dim Multiple Integral

Variable Substitution in n -dim Multiple Integral

$$\int_I f(x^1, \dots, x^n) dx^1 \dots dx^n$$

$$= \left[\begin{array}{l} x^1 = x^1(\xi^1, \dots, \xi^n) \\ x^2 = x^2(\xi^1, \dots, \xi^n) \\ \dots \\ x^n = x^n(\xi^1, \dots, \xi^n) \end{array} \right] \quad /parameterize/$$

$$= \int_{\Delta} f(x^1(\xi^1, \dots, \xi^n), \dots, x^n(\xi^1, \dots, \xi^n)) \left| \frac{D(x)}{D(\xi)} \right| d\xi^1 \dots d\xi^n$$

where $\frac{D(x)}{D(\xi)} = \frac{D(x^1, \dots, x^n)}{D(\xi^1, \dots, \xi^n)}$ is the Jacobian determinant of the transformation.

4 Multiple Integrals

4.1 The Riemann Integral over an n -Dimensional Interval

4.1.1 Comprehensions

First, the definition of Riemann integral over n -dimension interval is simimiar (almost same) with 1-dimension condition ($\int_I f(x)dx := \lim_{\lambda(P) \rightarrow 0} \sigma(f, P, \xi)$),

$$\int_I f(x^1, \dots, x^n) dx^1 \dots dx^n \text{ or } \int \dots \int_I f(x^1, \dots, x^n) dx^1 \dots dx^n$$

Then, introduce the necessity condition of it ($f \in \mathcal{R}(I) \implies f$ is bounded on I). After that, book gives some properties (Lemmas) about the set of measure zero (same with 1-dimension condition), and developes the Cantor Theorem (about uniform continuity of continuous functions on compact set). At last, we're given 2 criterions (necessity and sufficiency) of the Riemann integral (Darboux Criterion, Lebesgue Criterion).

Example. Let $f : I \rightarrow \mathbb{R}$ is a continuous real-value function on $I \subset \mathbb{R}^{n-1}$, then it's graph is a set measure zero in n -dimension.

Lemma 4.1. /Development of Cantor Theorem/ If the relation $\omega(f; x) < w_0$ holds at each point of a compact set K for the function $f : K \rightarrow \mathbb{R}$, then for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\omega(f; U_\delta(x)) < w_0 + \varepsilon$ for each point $x \in K$.

Theorem 4.1. (Lebesgue's criterion). $f \in \mathcal{R}(I) \Leftrightarrow (f \text{ is bounded on } I) \wedge (f \text{ is continuous almost everywhere on } I)$.

Theorem 4.2. (Darboux Criterion) $f \in (I) \Leftrightarrow (f \text{ is bounded on } I) \wedge (\underline{\mathcal{J}} = \overline{\mathcal{J}})$, where $\underline{\mathcal{J}} = \sup_P s(f, P)$, $\overline{\mathcal{J}} = \inf_P S(f, P)$.

4.1.2 Problems Sets

Problem 1. a) Show that a set of measure zero has no interior points.

- b) Show that not having interior points by no means guarantees that a set is of measure zero.
- c) Construct a set having measure zero whose closure is the entire space $\mathbb{R}^n$.
- d) A set $E \subset I$ is said to have content zero if for every $\varepsilon > 0$ it can be covered by a finite system of intervals $I_1, \dots, I_k$ such that $\sum_{i=1}^k |I_i| < \varepsilon$. Is every bounded set of measure zero a set of content zero?
- e) Show that if a set $E \subset \mathbb{R}^n$ is the direct product $\mathbb{R} \times e$ of the line $\mathbb{R}$ and a set $e \subset \mathbb{R}^{n-1}$ of $(n-1)$ -dimensional measure zero, then E is a set of n -dimensional measure zero.

Proof. a) [Proof by contradiction] If E has a point x_0 as an interior point, then there exists a ball $B(x_0, r)$ such that $B(x_0, r) \subset E$. Since the ball has positive measure, it follows that E cannot have measure zero.

b) [Counterexample] Consider the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$, which have no interior points in $\mathbb{R}$ but are not of measure zero.

c) /inspired by b)/ Consider the set $E = \mathbb{Q}^n$, the set of all points in $\mathbb{R}^n$ with rational coordinates. The closure of E is $\mathbb{R}^n$, and E has measure zero since it can be covered by countably many intervals whose total length can be made arbitrarily small.

d) [Principle] No, since the Lebesgue measure zero set can be covered by at most countable intervals (which means measure $\mu = o(1) \cdot \mathcal{O}(1)$), while a set of content zero can be covered by a finite number of intervals (which means the content (also a kind of measure) sometimes is $\mathcal{O}(1) \cdot \mathcal{O}(1)$). [Example] The $\mathbb{Q} \cap [0, 1]$ and Cantor set are sets of measure zero but none of content zero.

e) From the construction of the product measure λ_n (Lebesgue measure of $\mathbb{R}^n$) /Lemma1/ if $A \in \mathcal{B}(\mathbb{R}^k)$ and $B \in \mathcal{B}(\mathbb{R}^{n-k})$, ($0 < k < n$), then $\lambda_n(A \times B) = \lambda_k(A)\lambda_{n-k}(B)$. So here (using also that /Lemma2/ if $E_m \uparrow E$, $\lim_{m \rightarrow +\infty} \lambda_n(E_m) = \lambda_n(E)$) $\lambda_n(E) = \lambda_n(\mathbb{R} \times e) = \lim_{m \rightarrow +\infty} \lambda_n((-m, m) \times e) = \lim_{m \rightarrow +\infty} \lambda_1((-m, m)) \times \lambda_{n-1}(e) = \lim_{m \rightarrow +\infty} \lambda_1((-m, m)) \times 0 = 0$.

Notes. 1. The proof of e) /Lemma2/ could found in *Real and Complex Analysis* by Walter Rudin, Theorem 1.19(Page 16), based on the (positive) measure definition (countable additive of disjoint countable measurable sets).

- Problem 2.** a) Construct the analogue of the Dirichlet function in $\mathbb{R}^n$ and show that a bounded function $f : I \rightarrow \mathbb{R}$ equal to zero at almost every point of the interval I may still fail to belong to $\mathcal{R}(I)$.
- b) Show that if $f \in \mathcal{R}(I)$ and $f(x) = 0$, at almost all points of the interval I , then $\int_I f(x) dx = 0$.

Proof. a) [Counterexample] Consider $\mathcal{D}^n = \begin{cases} 1 & \text{if } x \in \mathbb{Q}^n \\ 0 & \text{if } x \in \mathbb{R}^n \setminus \mathbb{Q}^n \end{cases}$, the discontinuous set is $\mathcal{R}^n$.

b) /by definition/ $\int_I f(x) dx = \lim_{\lambda(P) \rightarrow 0} \sum_{\xi_i \in P} f(\xi_i) \Delta(x_i) = \lim_{\lambda(P) \rightarrow 0} \sum_{\xi_i \in A \cap P} f(\xi_i) \Delta(x_i) + \lim_{\lambda(P) \rightarrow 0} \sum_{\xi_j \in B \cap P} f(\xi_j) \Delta(x_i) = \mathcal{O}(1) \cdot o(1) + o(1) \cdot \mathcal{O}(1)$, where $A = \{\cup [x_k, x_{k+1}] \subset P : \forall k, \exists \xi \in [x_k, x_{k+1}], \text{ s.t. } f(\xi) \neq 0\}$ and $B = I \setminus A$.

Problem 3. There is a small difference between our earlier definition of the Riemann integral on a closed interval $I \subset \mathbb{R}$ and Definition 7 for the integral over an interval of arbitrary dimension. This difference involves the definition of a partition and the measure of an interval of the partition. Clarify this nuance for yourself and verify that

$$\int_a^b f(x) dx = \begin{cases} \int_I f(x) dx & \text{if } a < b \\ - \int_I f(x) dx & \text{if } a > b \end{cases}$$

where I is the interval on the real line $\mathbb{R}$ with endpoints a and b .

Proof. Just understand it, it's not hard.

- Problem 4.** a) Prove that a real-valued function $f : I \rightarrow \mathbb{R}$ defined on an interval $I \subset \mathbb{R}^n$ is integrable over that interval if and only if for every $\varepsilon > 0$ there exists a partition P of I such that $S(f; P) - s(f; P) < \varepsilon$.
- b) Using the result of a) and assuming that we are dealing with a real-valued function $f : I \rightarrow \mathbb{R}$, one can simplify slightly the proof of the sufficiency of the Lebesgue criterion. Try to carry out this simplification by yourself.

Proof. a) /by The Darboux Criterion/ $f \in (I) \Leftrightarrow (f \text{ is bounded on } I) \wedge (\underline{\mathcal{J}} = \overline{\mathcal{J}})$, where $\underline{\mathcal{J}} = \sup_P s(f, P)$, $\overline{\mathcal{J}} = \inf_P S(f, P)$. Notice that $\forall \varepsilon > 0, \exists P, \text{ s.t. } S(f, P) - s(f, P) < \varepsilon \Leftrightarrow \underline{\mathcal{J}} = \overline{\mathcal{J}}$ (proof is similar with 1-dimension condition), which is what we need to prove.

b) [Segmented estimation] Sufficiency: procedure is same with book, a) just simplify the last step.

4.2 The Integral over a Set

4.2.1 Comprehensions

This part is not very long and mainly revolves around the concept of "admissible set".

First, the book introduces the concept (i.e. definition) of "admissible set" and some (operational) properties of it. Then, using a characteristic function χ_E of set E , we're given the definition of the integral over a set E and the corresponding Lebesgue's Criterion (over "admissible set"). At last, the book also gives the Jordan measure (or volume) of a bounded set E (actually it's only well defined for "admissible set").

Definition 1. An admissible set is a bounded set $E \subset \mathbb{R}^n$ s.t. ∂E is measure zero in sense of Lebesgue.

Lemma 4.2. $\forall$ sets $E, E_1, E_2 \subset \mathbb{R}^n$,

- a) ∂E is the closed set on $\mathbb{R}^n$.
- b) $\partial(E_1 \cup E_2), \partial(E_1 \cap E_2), \partial(E_1 \setminus E_2) \subset \partial E_1 \cup \partial E_2$.
- c) Finite union or intersection of admissible sets is still admissible set, the difference of admissible sets is still admissible set.

Notes. 1. Generally, for infinite admissible sets, Lemma 2.2 c) is not true.
2. The boundary of admissible set is compact on $\mathbb{R}^n$ (closed and bounded).

Definition 2. The integral of f over E is given by $\int_E f(x)dx := \int_{I \supset E} f\chi_E(x)dx$, where I is any interval containing E . If the integral exists, we say $f \in \mathcal{R}(E)$.

$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$ is the characteristic function of set E and $f_{\chi_E}(x) = \begin{cases} f(x) & x \in E \\ 0 & x \notin E \end{cases}$.

Notes. 1. $\chi_E(x)$ is interrupted only on the boundary ∂E . The points of discontinuity of $f_{\chi_E}(x)$ are either points of discontinuity of f on E , or the result of discontinuities of χ_E , in which case they lie on ∂E .
2. $\forall I_1, I_2 \supset E$, integrals $\int_{I_1} f_{\chi_E}(x)dx$ and $\int_{I_2} f_{\chi_E}(x)dx$ either both exist (then values are equal) or both fail to exist.

Theorem 4.3. A function $f : E \rightarrow \mathbb{R}$ is integrable over an admissible set if and only if it is bounded and continuous at almost all points of E .

Notes. 1. Compared with f , f_{χ_E} may have additional points of discontinuity only on the ∂E .

Definition 3. The (Jordan) measure or content of a bounded set $E \subset \mathbb{R}^n$ is $\mu(E) := \int_E 1 \cdot dx = \int_{I \supset E} \chi_E(x) dx$, provided this Riemann integral exists.

Notes. 1. Only admissible sets, are measurable in the sense of aforementioned definition.

Comprehensions for Definition 3 (From Book.)

The geometric meaning of $\mu(E)$ (if E is admissible) could be seen in following way:

$$\mu(E) = \int_{I \supset E} \chi_E(x) dx = \int_{\underline{I \supset E}} \chi_E(x) dx = \overline{\int_{I \supset E} \chi_E(x) dx}$$

Due to Darboux Criterion and Theorem, $\mu(E)$ is the common limit as $\lambda(P) \rightarrow 0$ of the volumes of polyhedra inscribed in and circumscribed about E , in agreement with the accepted idea of the volume of simple solids $E \subset \mathbb{R}^n$.

Why the $\mu(E)$ is called Jordan measure?

Definition 4. A set $E \subset \mathbb{R}^n$ is a set of measure zero in the sense of Jordan or a set of content zero if for every $\varepsilon > 0$ it can be covered by a finite system of intervals $I_1, \dots, I_k$ s.t. $\sum_{i=1}^k |I_i| < \varepsilon$.

Compared with measure zero in the sense of Lebesgue, a requirement that the covering be finite appears here, shrinking the class of sets of Lebesgue measure zero. For example, the set of rational points is a set of measure zero in the sense of Lebesgue, but not in the sense of Jordan.

In order for the least upper bound of the contents of polyhedra inscribed in a bounded set E to be the same as the greatest lower bound of the contents of polyhedra circumscribed about E (and to serve as the measure $\mu(E)$ or content of E), it is obviously necessary and sufficient that the boundary ∂E of E have measure 0 in the sense of Jordan. That is the motivation for the following definition.

Definition 5. A set E is Jordan-measurable if it is bounded and its boundary ∂E has Jordan measure zero.

Hence, the class of Jordan-measurable subsets is precisely the class of admissible sets. That's the reason the measure $\mu(E)$ defined earlier can be called (and is called) the Jordan measure of the (Jordan-measurable) set E .